

Government-Backed Aquaculture Equipment Suppliers and Technology Procurement Frameworks in India

Policy Context and Institutional Architecture

The structural transformation of the Indian fisheries and aquaculture sector from traditional capture methods to highly intensive, technology-driven culture systems is anchored by comprehensive public sector interventions. Operating under the Department of Fisheries within the Ministry of Fisheries, Animal Husbandry & Dairying, the National Fisheries Development Board (NFDB) serves as the primary administrative catalyst for technology infusion, capacity building, and infrastructure modernization across India.

To bridge systemic gaps in productivity, biosecurity, and cold-chain logistics, the government has deployed key policy instruments, most notably the Pradhan Mantri Matsya Sampada Yojana (PMMSY) with an estimated investment of Rs. 20,050 crore.¹ Accessing these benefits requires registration on the National Fisheries Digital Platform (NFDP), which streamlines aquaculture insurance, performance grants, credit facilitation, and supply-chain traceability.

Central to the implementation of both PMMSY and the Fisheries and Aquaculture Infrastructure Development Fund (FIDF) is the empanelment of equipment manufacturers and suppliers. By maintaining a rigorous registry of approved vendors, the NFDB and state fisheries directorates standardize technology deployments, establish transparent benchmark costs, and ensure that public subsidy outlays are directed toward durable, high-performance assets.

Standardized Aquaculture Equipment and Financial Assistance

Under government-backed frameworks, the financial viability of modernizing an aquaculture enterprise is highly dependent on capital subsidies. To guide procurement, the government establishes benchmark unit costs for infrastructure and equipment.³ These costs dictate the maximum permissible financial assistance, which typically ranges from 40% for general category individuals to 60% for scheduled castes, scheduled tribes, and women entrepreneurs under Centrally Sponsored Schemes.

Equipment / Project Category	Standardized Unit Cost	Funding & Operational Scope under Government Schemes
Establishment of Fishing Harbours	Rs. 15,000.00 Lakhs	Infrastructure development for marine capture fisheries. ³

Establishment of Aquatic Quarantine Facilities	Rs. 2,500.00 Lakhs	Biosecurity infrastructure for pathogen-free seed imports. ³
Establishment of Fish Processing Units	Rs. 4,674.00 Lakhs	Value-addition and export-oriented processing facilities. ³
Setting up of Brood Banks	Rs. 1,000.00 Lakhs	Certified genetic resource development for hatcheries. ³
Modernization of State Fish Seed Farms	Rs. 500.00 Lakhs	Infrastructure upgrades for high-quality seed production. ³
Integrated Cold Chain Facilities	Rs. 500.00 Lakhs	Post-harvest preservation to minimize economic losses. ³
Construction of Cold Storages / Ice Plants	Rs. 100.00 Lakhs	Marine and inland fisheries temperature-controlled storage. ³
Fish Feed Mills / Plants (10 Tonne/Day)	Rs. 650.00 Lakhs	Large-scale commercial nutritionally balanced feed production. ³
Fish Feed Mills / Plants (4-5 Tonne/Day)	Rs. 10.00 Lakhs	Small-to-medium scale localized feed manufacturing. ³
Deep Sea Fishing Vessels	Rs. 80.00 Lakhs	Technological upgradation of marine capture craft. ³
Development of Hatcheries	Rs. 50.00 Lakhs	Breeding facilities for freshwater and marine species. ³
Establishment of Disease Diagnostic Labs	Rs. 150.00 Lakhs	Localized pathology and water quality testing facilities. ³

Development of Aquaculture (per Hectare)	Rs. 7.00 Lakhs	Land-based grow-out pond construction and inputs. ³
Sea Cage Culture (per Unit)	Rs. 5.00 Lakhs	Open-water marine grow-out cage systems. ³
Mariculture Nursery Area (per Hectare)	Rs. 6.00 Lakhs	Near-shore coastal seed rearing infrastructure. ³
Cage Culture in Reservoirs	Rs. 3.00 Lakhs	Inland open-water cage arrays. ³
Air Blowers (10 HP)	Rs. 1,20,000	Subsidized rate; maximum subsidy of Rs. 48,000. ⁴
Air Blowers (5 HP)	Rs. 70,000	Subsidized rate; maximum subsidy of Rs. 28,000. ⁴
Paddle Wheel Aerators (2 HP)	Rs. 40,000	Subsidized rate; maximum subsidy of Rs. 16,000. ⁴
Paddle Wheel Aerators (1 HP)	Rs. 25,000	Subsidized rate; maximum subsidy of Rs. 10,000. ⁴
Fishing Nets with Fittings (300 Ft)	Rs. 50,000	Subsidized rate; maximum subsidy of Rs. 20,000. ⁴
Bio-Security Nets	Rs. 20,000	Subsidized rate; maximum subsidy of Rs. 8,000. ⁴
Insulated Box (500 Liter Capacity)	Rs. 15,000	50% subsidy available for up to 3 units per applicant. ⁵
Rearing Space Development (per Hectare)	Rs. 4,00,000	Maximum allowable expenditure for nursery pond renovation. ⁵

National and Regional Empaneled Supplier Registry

To streamline procurement under subsidized categories, the NFDB and the Coastal Aquaculture Authority (CAA) maintain a validated database of empaneled manufacturers and service

providers. The registry consists of 41 officially empaneled equipment manufacturers and suppliers. These companies are subject to rigorous evaluations and are listed with exact addresses, contact numbers, and corporate credentials to allow direct farmer-to-vendor communication.

Supplier Name & Brand	Exact Web Source & Document Link	Address & Coordinates	Approved Products & Scope	Phone Number & Email Contacts	Validity & Status
M/s I.A.I.A. General Supplier	(https://www.scribd.com/document/721741141/Web-Template-I-A-I-A-General-Supplier) ⁶	Rajiv Complex, 11-12, Near Bus Stand, Nagri, Dhamtari, Chhattisgarh - 493778 ⁶	Sea cages, HDPE/GI/pontoon cages (HMW HDPE modular), UV-resistant pen cages, cage nets (HDPE/polyester/PPDy Neema), FRP boats, and floating jetties ⁶	Phone: +91 9425223365 Email: khanasif900@gmail.com Rep: Mr. Asif Iqbal Khan ⁶	Valid up to July 31, 2024 ⁶
M/s Commercial Syn Bags Limited (COMSYN)	(https://www.scribd.com/document/721741560/Web-Template-Commercial-Syn-Bags-Limited) ⁷ ; (https://comsyn.com/)	Registered Office: Commercial House, 3-4, Jaora Compound, M.Y.H. Road, Indore - 452001, Madhya Pradesh, India	Low-density polyethylene (LDPE) pond liners, geomembranes, geotextile fabrics for waterproof pond lining, heavy-duty tarpaulins, FIBC bags, and food-grade	Phone: +91-731-4279525 / 26; +91-07292415151 / 152 Email: comsyn@comsyn.com; mails@comsyn.com; cs@comsyn.com	Valid up to July 31, 2024 ⁹

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Dhanashree Solar Energy Solution	(https://midc.directory.com/ad/dhanashree-solar-energy-solution/) ¹⁰ ; (https://www.dhanashreesolar.net/about.html) ¹¹	MIDC Area, Thane, Maharashtra, India	Solar RO water purifiers, solar-powered freezers, solar heat pump dryers, solar-powered marine and submersible pumps, solar cold storage units, and solar water chillers ¹⁰	Phone: +91 7972499093 Email: response@dhanashreesolar.net; dhanashreesolarenergy@gmail.com ¹⁰	Active; recognized as an empanelled green energy equipment vendor by NFDB ¹⁰
Jamshedpur Resources Private Limited (JRPL / Duraplast)	(https://www.indiamart.com/jamshedpur-resources/profile.html) ¹² ; (https://5.imimg.com/data5/SELLER/Doc/2025/10/552265376/YT/WE/YT/15675029/floating-structures-platform.pdf) ¹³	Jamshedpur - 831002, Jharkhand, India ¹²	HDPE fish cages for inland cage culture, modular HDPE floating pontoons, floating jetties, and walkways under Blue Revolution/ PMMSY ¹³	Phone: +91 8047641967 Web: www.duraplast.org.in CEO: Mr. Ravi Kejriwal ¹²	Active; CIPET-tested, NSIC-certified, and empanelled with NFDB ¹²
RK and Sons	(https://www.rkandsons.co.in/) ¹⁴	Head Office: Chhattisgarh, India	Floating Venturi systems, paddle	Phone: +91 99813 13066; +91 98939	Active; independent private partner

		(Serving Pan-India) ¹⁴	wheel aerators, Biofloc/RAS tanks, water pumps, HDPE pipes, and certified seeds ¹⁴	07540; +91 99810 34763 Email: info@rkandsons.co.in ¹⁴	serving 25+ states ¹⁴
M/s. Shrimp Improvement Systems LLC	<u>Coastal Aquaculture Authority Official Letters</u>	88005 Overseas Highway, 10-166, Islamorada, Florida - 33036, USA	Specific Pathogen-Free (SPF) Litopenaeus vannamei shrimp broodstock and post-larvae	Web: caa.gov.in (Overseas registry inquiries) Email: caaheadoffice@caa.gov.in	Active; empanelment period valid up to March 17, 2029
M/s. Shrimp Improvement Systems Hawaii, LLC	<u>Coastal Aquaculture Authority Official Letters</u>	73-4460 Queen Kaahumanu Highway, Suite - 108, Kailua-Kona, Hawaii - 96740, USA	SPF Litopenaeus vannamei shrimp broodstock and post-larvae	Web: caa.gov.in Email: caaheadoffice@caa.gov.in	Active; empanelment period valid up to March 17, 2029
M/s. Konabay, Sunrise Capital, Inc. (dba Kona Bay Marine Resources)	<u>Coastal Aquaculture Authority Official Letters</u>	P.O. Box No.1282, Kekaha, Kauai, Hawaii - 96752, USA	SPF Litopenaeus vannamei shrimp broodstock and post-larvae	Web: caa.gov.in Email: caaheadoffice@caa.gov.in	Active; empanelment period valid up to March 17, 2029

Portal Booking Functionality and Procurement Framework

A critical administrative aspect of government-backed aquaculture lists is understanding how transactions are initiated. Many operators seek a direct booking or e-commerce engine on government websites.⁶

Absence of Direct Booking Portals

Neither the official **National Fisheries Development Board (NFDB)** portal (nfdb.gov.in) nor the central **National Fisheries Digital Platform (NFDP)** portal (nfdp.dof.gov.in)⁹ contains a direct booking engine, e-commerce checkout, add-to-cart, or transaction processing gateway to purchase or lease equipment.⁶ The portals are designed strictly as **administrative registries and directory-lookup databases**.

The Standard Procurement Flow

Because the portals lack booking engines, the process of procuring subsidized aquaculture equipment from these registries requires the following independent offline steps:

1. **Digital Registration & ID Generation:** The buyer must first complete their registration on the National Fisheries Digital Platform (NFDP) website (nfdp.dof.gov.in)⁹ to obtain a work-based digital identity.
2. **Directory Browsing:** The buyer reviews the list of 41 empanelled equipment manufacturers and suppliers published on the NFDB website.
3. **Direct Contact and RFQ:** The buyer must independently establish contact with the empanelled manufacturer using their provided coordinates (e.g., calling M/s I.A.I.A. General Supplier at +91 9425223365⁶ or email COMSYN at comsyn@comsyn.com).
4. **Offline Quotation & Negotiation:** The buyer requests a written quotation as per standard technical specifications.⁶ The commercial terms, payment methods (e.g., cash, bank transfer, letter of credit), and timeline negotiations are conducted directly between the buyer and vendor, entirely outside the scope of the government portal.⁶
5. **Subsidy Reimbursement Claims:** Once a purchase is finalized, the invoice, proof of payment (DBT transfer), and equipment installation pictures are submitted along with land/lease documents to the Block or District Fisheries Officer to claim the 40% to 60% PMMSY subsidy reimbursement.⁶

Technical Engineering Standards for Subsidized Infrastructure

To safeguard public investments, the government mandates strict material and structural specifications for critical aquaculture assets.¹⁶ This is highly evident in open-water cage farming, where structures must endure extreme physical stress from waves, currents, and biological fouling. Working in coordination with the Central Marine Fisheries Research Institute (CMFRI), the NFDB enforces specific structural criteria for floating marine and reservoir installations.¹⁶

HDPE Floating Cage Frame Assembly

The physical frame forms the structural foundation of the cage system and must be fabricated to

precise dimensional tolerances using certified engineering materials.

- **Sizing and Geometry:** Cages must feature a circular floating configuration with a standardized inner diameter of 6 meters.
- **Collar Construction:** Double floating collar design using PE100 virgin-grade, UV-stabilized HDPE pipe. The outer diameter of the floating pipes must be 140 mm with a wall thickness of 10 mm to 12 mm.
- **Railing Assembly:** Handrail height must stand exactly 1 meter above the floating collars. The railing must be constructed from 75 mm outer diameter PE100 virgin HDPE pipe with a wall thickness of 8 mm to 10 mm.
- **Structural Support:** The structural loop must be reinforced with nine heavy-duty, injection-moulded HDPE brackets. These brackets must have a matching inner diameter of 140 mm and be secured along the collars using robust physical stoppers to prevent slippage.

Specialized Aquaculture Netting Systems

Netting configurations are species- and stage-specific, designed to optimize water exchange while preventing escapes and predator intrusion.

Net Type / Purpose	Material Specification	Structural Dimensions	Mesh Size & Knotting
Predator Net	HDPE Knotted 280D, 63-ply	7 m Diameter x 5 m Depth	80 mm knot-to-knot mesh
Grow-Out Net	HDPE Knotted 280D, 63-ply	6 m Diameter x 5 m Depth	40 mm knot-to-knot mesh
Fingerling Net	HDPE Knotted 280D, 63-ply	6 m Diameter x 5 m Depth	20 mm knot-to-knot mesh
Bird Net	HDPE Knotted 280D, 24-ply	6 m Diameter	100 mm knot-to-knot mesh

To ensure that the netting chambers maintain their proper cylindrical volume under the influence of strong ocean or river currents, the entire structure must be weighted. The specification

demands a minimum of 16 concrete ballast blocks, weighing 10 kg each, connected to the lower net rim using 10 mm diameter mild steel rings.

Heavy-Duty Mooring and Anchoring Sub-Systems

The mooring array stabilizes the floating structure, absorbing kinetic energy from wave actions and preventing displacement.

- **Primary Anchors:** Dual 100 kg mild steel anchors positioned to counteract prevailing currents.
- **Ground Chains:** Two 5 meter segments of 18 mm heavy-duty mild steel chain acting as catenary weight to keep anchor pull horizontal.
- **Rope Assembly:** High-tensile, UV-stabilized 24 mm diameter HDPE anchor ropes extending up to 60 meters in length.
- **Rigging Hardware:** Ropes must be fitted with protective metal thimbles, heavy-duty rope clips, and certified 12 Ton capacity D-shackles to prevent wear at critical junction points.
- **Navigation and Safety:** Compliance with maritime safety laws is maintained by integrating dual 10 L marker buoys, robust 10 mm HDPE marker ropes, and an automated solar-powered marker light.

Strategic Evaluation of Global vs. Domestic Supplier Networks

Aquaculture operators face a critical choice between procuring through the domestic, government-subsidized NFDB-empaneled ecosystem and sourcing from international markets. International directories, such as the US-based National Aquaculture Association (NAA), highlight global manufacturers specializing in highly advanced, automated machinery.⁷ However, these international suppliers are not covered under Indian subsidy programs.⁶

Global Supplier Name	Geographic Location	Specialized Aquaculture Equipment	Operational Context & Market Role
Norfab Equipment Ltd.	West Scotland, United Kingdom ¹⁹	Bespoke fish farming equipment, versatile freestanding hydraulic cranes, and customized engineering systems. ¹⁹	International engineering firm supplying high-end bespoke systems to global salmon, barramundi, and shrimp operations. ¹⁹
Fuel Tech, Inc.	Illinois, United States ⁷	Innovative water treatment	Large-scale environmental

		technologies, commercial-scale chemical feeding, and emissions control systems. ⁷	technology provider servicing industrial-grade water conditioning plants. ⁷
Fusion Fiberglass Products	Florida, United States ⁷	Custom-molded fiberglass reinforced plastic (FRP) holding and breeding tanks. ⁷	Precision manufacturer of specialized bio-secure tanks for hatcheries and recirculating aquaculture systems (RAS). ⁷
Fiberglass Creations	Alabama, United States ⁷	Industrial-grade fiberglass holding tanks and water treatment vessels. ⁷	Specialized engineering shop supplying non-corrosive structural containment units. ⁷
YSI (A Xylem Brand)	Ohio, United States ¹⁸	High-precision dissolved oxygen sensors, multi-parameter water quality probes, and continuous monitoring arrays. ¹⁸	Industry-standard scientific instrumentation provider for commercial hatcheries and high-density RAS. ¹⁸
Kasco Marine	Wisconsin, United States ¹⁸	Submersible aerators, surface de-icers, water circulators, and decorative fountain aerators. ¹⁸	Specialized engineering firm focusing on pond destratification, water circulation, and active aeration. ¹⁸
OxyGuard International	Farum, Denmark ¹⁸	Automated water quality measuring systems, stationary	Specialized bio-monitoring firm providing

		gas monitoring, and alarm systems. ¹⁸	computerized command-and-control systems for modern super-intensive farms. ¹⁸
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While international manufacturers offer high-precision monitoring, sensor automation, and advanced metallurgy, their integration into Indian operations is typically limited to large-scale corporate ventures that do not rely on public subsidies. For smallholders, cooperative societies, and regional entrepreneurs, local empanelment remains the only pathway to securing the 40% to 60% financial assistance necessary to offset high initial capital requirements.

Decentralized State-Level Governance: The Bihar Model

National policies are operationalized through decentralized state fisheries directorates, which customize empanelments to fit regional hydrology and socioeconomic conditions. The state of Bihar provides an excellent case study of how localized procurement frameworks can accelerate production.⁶ By tailoring technology empanelments, the Directorate of Fisheries in Patna has driven Bihar's annual fish production up to 8.73 lakh metric tonnes, placing the state fourth nationally in inland fish production.²⁰

Bihar's procurement and development strategies focus on three key technological interventions:

- **Aquaculture Solarization:** Addressing high operational costs is a major challenge for inland pond operators. Bihar has implemented Expression of Interest (EOI) programs to empanel specialized engineering firms for installing boring-cum-solar submersible pumpsets, shifting rural farms away from costly diesel-powered pumping.
- **Prefabricated Fresh Catch Kiosks:** To improve post-harvest hygiene and urban supply chain logistics, the state empanels fabrication agencies to build, supply, and install modular, temperature-controlled retail booths called "Fresh Catch Kiosks" alongside mobile lab-cum-awareness vans.
- **Local Craft and Gear Schemes:** Under targeted welfare programs, district-level administrations (such as Vaishali) empanel local boatbuilders and net weavers to distribute packages of wooden and FRP boats alongside high-quality cast nets directly to riverine fishers and local cooperative societies.⁶

Legal Jurisdictions, Risk Mitigation, and Procurement Governance

Procuring specialized machinery through government-backed listings requires a clear understanding of the regulatory and legal frameworks established by the NFDB.⁶ The relationship between the board, the empaneled manufacturer, and the purchasing entity is defined by strict liability limitations.⁶

Liability Allocations and Legal Disclaimers

The NFDB serves strictly as an administrative facilitator. It maintains clear boundaries regarding transactions initiated through its portal ⁶:

- **No Quality Guarantees:** The board does not guarantee, certify, or assume liability for the physical quality, performance, or structural longevity of the equipment supplied by empaneled vendors.⁶
- **Zero Transactional Liability:** The NFDB is completely insulated from financial agreements.⁶ It assumes no responsibility for payment disputes, transaction delays, or defaults between buyers and suppliers.⁶
- **Accuracy of Portal Data:** The responsibility for ensuring the accuracy of pricing, material grades, and regulatory compliance on the portal rests entirely with the listed seller.⁶
- **Exclusive Jurisdictional Limits:** Any legal disputes arising from contract defaults, manufacturing flaws, or failure to deliver must be settled in courts within the specific local jurisdiction where the contract was executed.⁶ The NFDB cannot be named as a party in such disputes.⁶

Recommended Risk Mitigation Protocols for Buyers

To minimize operational and financial exposure when procuring through government listings, purchasing entities should establish independent verification protocols:

1. **Independent Pre-Dispatch Inspections:** Before final dispatch, buyers should conduct third-party technical audits to verify raw material compliance, such as testing the thickness and density of HDPE pipes or checking UV-resistance certificates for geomembranes.
2. **Performance-Linked Escrow Accounts:** Rather than releasing full payments upfront, structure payments around milestones, retaining a percentage (e.g., 15% to 20%) until the system has run successfully for a set trial period.⁶
3. **Direct Service-Level Warranties:** Secure direct, legally binding manufacturer warranties (minimum of 12 months for cage frames and aeration motors) that operate independently of the NFDB listing.

By combining government financial subsidies with proactive risk management, aquaculture operators can confidently deploy modern technology, driving sustainable productivity gains while protecting their capital investments.

Works cited

1. Content - Department of Fisheries, accessed on May 20, 2026, <https://dof.gov.in/static/uploads/2025/08/f6cd5364273875538832671fa2e0b965.pdf>
2. Pradhan Mantri Matsya Sampada Yojana | Department of Fisheries, MoCIT, GoI, accessed on May 20, 2026, <https://www.pmsy.dof.gov.in/>
3. FISHERIES AND AQUACULTURE INFRASTRUCTURE DEVELOPMENT FUND

- (FIDF) ONLINE PORTAL, accessed on May 20, 2026, <https://www.fidf.in/>
4. Fishery Machinery Subsidy Scheme 2021-22 | PDF - Scribd, accessed on May 20, 2026, <https://www.scribd.com/document/752451603/1-Subsidy-for-Fisheries-Machinery-Equipments-for-Intensive-Aquaculture>
 5. Fisheries Sector Support Schemes in India 2019-2020 Compiled by ICSF, accessed on May 20, 2026, https://www.icsf.net/wp-content/uploads/2022/06/Fisheries_Support_Schemes_2018-2019.pdf
 6. I.A.I.A. Supplier for Fisheries Equipment | PDF | Supply Chain | Business - Scribd, accessed on May 20, 2026, <https://www.scribd.com/document/721741141/Web-Template-I-A-I-A-General-Supplier>
 7. Empanelled Fisheries Equipment Suppliers | PDF | Supply Chain, accessed on May 20, 2026, <https://www.scribd.com/document/721741560/Web-Template-Commercial-Syn-Bags-Limited>
 8. Pond Liner Suppliers & Manufacturers - Environmental XPRT, accessed on May 20, 2026, <https://www.environmental-expert.com/companies/keyword-pond-liner-14032>
 9. Home | National Fisheries Digital Platform - Department of Fisheries, accessed on May 20, 2026, <https://nfdp.dof.gov.in/>
 10. Dhanashree Solar Energy Solution - MIDC, accessed on May 20, 2026, <https://midcdirectory.com/ad/dhanashree-solar-energy-solution/>
 11. About Us - Dhanashree Solar Energy Solution, accessed on May 20, 2026, <https://www.dhanashreesolar.net/about.html>
 12. Jamshedpur Resources Private Limited - Manufacturer from Jamshedpur, India | About Us, accessed on May 20, 2026, <https://www.indiamart.com/jamshedpur-resources/profile.html>
 13. JRPL Corporate Brochure, accessed on May 20, 2026, <https://5.imimg.com/data5/SELLER/Doc/2025/10/552265376/YT/WE/YT/15675029/floating-structures-platform.pdf>
 14. RK and Sons, accessed on May 20, 2026, <https://www.rkandsons.co.in/>
 15. Notice inviting Applications For Empanelment of Consultants /Consulting Firms for Technical and Managerial support services in F - NCDC, accessed on May 20, 2026, <https://www.ncdc.in/documents/tender/1714010426Notice-for-Empanelment-of-Consultants-Consulting-Firms-for-technical-and-managerial-support-services-in-Fisheries-DSFV-by-NCDC-V3.pdf>
 16. EXPRESSION OF INTEREST FOR THE EMPANELMENT OF SEA CAGE FABRICATORS - Central Marine Fisheries Research Institute, accessed on May 20, 2026, <https://www.cmfri.org.in/uploads/tender/Expression%20of%20Interest%20for%20the%20empanelment%20of%20sea%20cage%20fabricatorsEOI.pdf>
 17. expression of interest for the empanelment of sea cage fabricators - Central

Marine Fisheries Research Institute, accessed on May 20, 2026,
<https://www.cmfri.org.in/uploads/tender/SeaCage.pdf>

18. Special Thanks to Our 2025 Auction Donors! - - National Aquaculture Association, accessed on May 20, 2026, <https://www.nationalaquaculture.org/auction-donors/>
19. Aquaculture Engineering Equipment suppliers to the Fish Farming Industry | Norfab Equipment, accessed on May 20, 2026, <https://www.norfab.co.uk/>
20. Special Scheme to Promote Fish Production in Bihar - PIB, accessed on May 20, 2026,
<https://www.pib.gov.in/PressReleaseDetailm.aspx?PRID=2083150@=3&lang=2>
21. District Fisheries Officer: List of Empaneled agencies for the supply of fishing wooden boats, fishing FRP boat and cast nets under the boat and package Distribution Scheme - Vaishali, accessed on May 20, 2026,
<https://vaishali.nic.in/notice/district-fisheries-officer-list-of-empaneled-agencies-for-the-supply-of-fishing-wooden-boats-fishing-frp-boat-and-cast-nets-under-the-boat-and-package-distribution-scheme/>